

Rishank Singh

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Professional Summary

Ph.D. candidate in Computer Science with a focus on transfer learning and foundation models for LLMs, specializing in NLP and speech technologies. Experienced in developing voice-based systems and applying LLMs to real-world healthcare challenges, including dementia detection. Over 6 years of combined experience in research and software engineering, with strengths in algorithm development, full-stack implementation, and cross-functional collaboration. Passionate about leveraging AI for human-computer interaction, cognitive health, and impactful scientific advancement.

Education

UNIVERSITY OF MASSACHUSETTS, BOSTON

EXP MAY '27

- Ph.D.: Computer Science
- Cumulative GPA: 3.96

UNIVERSITY OF MASSACHUSETTS, BOSTON

MAY '24

- Master of Science: Computer Science
- Cumulative GPA: 4.0

RAJASTHAN TECHNICAL UNIVERSITY, JAIPUR

MAY '17

- Bachelor of Science: Computer Science

Research Experience

RESEARCH ASSISTANT | UMASS BOSTON |

MAR '23-PRESENT

- Conducting cutting-edge research on transfer learning under the LLM paradigm to build scalable and efficient models for various tasks.
- Collaborating on projects related to healthcare technology, including the detection of dementia using voice assistant systems and diet monitoring for older adults.
- **Developed a tracking device to monitor movements of geriatric patients, with a patent filed in 2023.** Presented findings at international conferences and published research in reputed journals.

Projects Undertaken:

- **Voice-Based LLM Chatbot for Dementia Detection:** Developed a conversational AI chatbot using Large Language Model (LLM) APIs, aimed at identifying cognitive decline in older adults by analyzing conversational patterns and linguistic features.
- **Movement Tracking Device for Physical Function Assessment:** Invented a wearable device to monitor and analyze the movement patterns of older adults, enabling objective evaluation of their physical functioning and mobility.
- **Alexa Command Integration for Cognitive Health:** Designed and implemented an application that seamlessly imports Alexa voice data to support early detection of dementia in older adults through speech analysis.
- **Diet Monitoring Dashboard for Older Adults:** Developed an interactive dashboard to track and manage dietary plans for older adults, supporting nutritional monitoring and personalized health interventions.

Teaching Experience

TEACHING ASSISTANT | UMASS BOSTON |

SEP '24-PRESENT

(CS-110 Introduction to Computing & CS-210 Intermediate Computing with Data Structures)

- Assisted with grading, lab sessions, and student support during office hours.
- Helped students troubleshoot code, explain programming concepts, and facilitated discussions.
- Supported course material preparation and provided feedback on assignments.
- Encouraged best coding practices and assist with projects and exams.

Professional Experience

STUDENT ASSISTANT | HEALEY LIBRARY, UMASS BOSTON |

APR '23-SEP '23

Projects Undertaken:

- **Library Subdomain Development:** Designed and implemented new web pages for Healey Library's subdomains, optimizing accessibility and user experience.
- **UMBrella Modernization:** Upgraded the legacy UMBrella application by integrating dynamic features using modern frameworks and technologies, enhancing performance and usability.
- **VPN Data Analytics:** Collected, processed, and visualized VPN usage data using Tableau, providing insights to improve network security and optimize resource allocation.

SOLUTION ENGINEER | NOKIA |

FEB '21 – JUL '22

Projects Undertaken:

- **Project management application:** Built an internal application to manage human and financial resources along with deadlines and productivity in a project.
- **Onboarding application:** Build an interactive react application for new joiners to the people and process of the firm.
- **Conference Application:** Part of team to build, manage and support teleconference meeting application.
- **IVR application:** Part of team to build, manage and support IVR application.

SOFTWARE ENGINEER | CHETU INC. |

JUL '17 – NOV '20

Projects Undertaken:

- **Jitsi conference application:** Software platform used for video communications (Meetings), messaging (Chat), voice calls (Phone)
- **Online video library application:** Online video library to maintain a healthy lifestyle at the time of the pandemic.
- **Human resource management application:** An application that could manage large-scale human resources in an organization
- **Mental health service application:** An application to ensure the mental health safety of an individual using online therapy.

Publications and Presentations

- Simone G Khandpekar, Rishank Singh, Caroline E Summerour, Jim Schwoebel, Xiaohui Liang, John A Batsis, *Feasibility and usability of a voice-based dietary recall tool in older adults: A pilot comparison with ASA-24*, SAGE Publications, Vol. 11, 2025.
- John A. Batsis, **Rishank Singh**, Jennifer Poole, Paige Bramblett, Danae Gross, Meredith Gilliam, Chaterlee Pamintuan, Beckham Nora, David H. Lynch, Xiaohui Liang, *A Preliminary Usability Evaluation of an Artificial Intelligence-based, motion-detecting Wearable device: the Geriatric Functional Assessment System*, Journal of Gerontology: Medical Sciences, 2025.
- **Singh, Rishank**; Zhu, Youxiang; Liang, Xiaohui; Batsis, John A.; and Summerour, Caroline, "SYMP25S: Can LLM detect dementia?" (2025). *Paul English Applied Artificial Intelligence (AI) Institute Publications*. 19.
- **Rishank Singh**, Sankalp Vaish, Xiaohui Liang, John Batsis, Jennifer Grace Poole, Danae Gross, *Clinical Geriatric Functional Assessment using Wearable Sensing and Machine Learning*, IEEE International Conference on E-health Networking, Application & Services, Healthcom 2024.
- **Rishank Singh**, Sahil Chaudhary, *Emotion Intelligence on Text Data Repositories*, SKIT Research Journal, Volume 6, Issue 1, 2016.

Skills

- Programming: Python, Java, TensorFlow, PyTorch, ReactJS, NextJS, JavaScript, C++
- Machine Learning & AI: Transfer learning, Foundation models, Large Language Models (LLMs), Deep learning frameworks
- Research Fields: Speech and language processing, Spoken language understanding, Multi-modal learning, In-context learning, Speech-based health diagnosis, Fine-grained image classification
- Data & Tools: Tableau, Visual Studio Code, IntelliJ IDEA, MySQL, PostgreSQL, MongoDB, Amazon RDS, SQLite

Recent awards and recognition

- Participated and completed at the new England Regional Innovation Node Sparks Program at MIT (Summer 2024).
- Filed a patent for a tracking device developed for geriatric patients at UMass Boston (2023).
- Awarded Best Project of the Quarter at Chetu Inc. (2018).
- Recognized for academic excellence and contributions to technology advancement at SKIT, Jaipur.